

Blood Bank Management System

Software Engineering Mini Project Part 1 Deliverables

Software Requirements Specification Test Plan and Software Architecture and Design Specification

Course UE24CS341A Software Engineering

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Submission Overview

This submission defines the requirements, planned verification, architecture and detailed design for a web-based Blood Bank Management System. The system covers donor registration, eligibility, appointments, donation recording, laboratory screening, blood-component inventory, hospital requests, cross-matching, issuance, alerts, reporting and audit logging.

The material is divided into three formal documents. Part A contains the Software Requirements Specification. Part B contains the Software Test Plan and detailed test cases. Part C contains the Software Architecture and Design Specification. Shared requirement identifiers provide traceability across all three parts.

Deliverables Included

- An IEEE-aligned SRS with descriptive sections, measurable functional and non-functional requirements, a UML use-case diagram and a dedicated security section.
- An IEEE-aligned Test Plan with scope, approach, environment, entry and exit criteria, security validation, traceability and nineteen detailed test cases.
- An IEEE-aligned Software Architecture and Design Specification with an architectural pattern, component diagram, security architecture, data design, three sequence diagrams, API design, error handling and requirement traceability.

Project Assumption

This document specifies an academic prototype. Clinical and statutory rules remain configurable and require approval from an authorized blood-centre professional before production use. The application supports decisions and record keeping; it does not replace clinical judgement.

Part A Software Requirements Specification

Document identifier BBMS SRS 1.0

1 Introduction

1.1 Purpose

This Software Requirements Specification defines the behaviour, interfaces, constraints and quality requirements of the Blood Bank Management System. It provides the baseline for design, implementation, testing and acceptance.

1.2 Scope

The system supports one or more blood-bank branches under one deployment. It maintains donor and donation records, applies configurable eligibility rules, tracks laboratory screening and component inventory, accepts hospital blood requests, records cross-matching and issuance, sends operational notifications and produces audit and management reports.

The first implementation may operate at one branch. The data model and access controls retain branch identifiers so that additional branches can be enabled without replacing the core design.

1.3 Intended Audience

The intended readers are the course evaluator, project team, student developers and student testers. Operational stakeholders include donors, blood-bank staff, hospital staff and system administrators.

1.4 Definitions and Abbreviations

Term	Meaning
API	Application Programming Interface
Blood unit	A uniquely identified whole-blood collection or derived component tracked by the system
Component	A blood product such as red cells, platelets or plasma
CSRF	Cross-Site Request Forgery
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
RTM	Requirements Traceability Matrix
SRS	Software Requirements Specification
UTC	Coordinated Universal Time

1.5 References

The normative and project references are listed in the References section at the end of this submission.

1.6 Document Conventions

The word shall indicates a mandatory and testable requirement. Functional requirements use FR identifiers, non-functional requirements use NFR identifiers and security requirements use SEC identifiers. Priorities indicate implementation importance for the academic prototype.

2 Overall Description

2.1 Product Perspective

The product is a role-based web application backed by a relational database. It centralizes records that may otherwise be kept in separate registers or spreadsheets. Browser clients communicate with the application over HTTPS. The application sends email through a configured provider and stores encrypted backups in restricted storage.

2.2 Product Functions

- Register donors and maintain consent, contact and medical-history information.
- Evaluate eligibility through configurable rules and active deferrals.
- Schedule donation appointments with branch and slot capacity controls.
- Record collection, component preparation, laboratory testing and unit state changes.
- Search inventory and manage hospital blood requests, reservations, cross-matches and issuance.
- Generate alerts, reports, backups and audit records.

2.3 User Classes

User class	Main responsibilities	Access level
Donor	Profile, eligibility, appointments and personal history	Own records only
Blood Bank Staff	Screening, donation, testing, inventory, requests and issue	Assigned branch operations
Hospital Staff	Availability search, request submission and request tracking	Hospital requests and restricted stock view
System Administrator	Accounts, roles, branches, configuration and reports	Administrative data without routine clinical editing

2.4 Operating Environment

- Server operating system: Linux.
- Application: Python and Django served by Gunicorn behind Nginx.
- Database: PostgreSQL through the Django object-relational mapper.
- Client: Supported desktop or mobile browser with JavaScript and cookies enabled.
- Deployment: Docker containers may be used, but containerization is not required for the academic prototype.

2.5 Design Constraints

- Server-side authorization is mandatory for protected operations.
- All persistent clinical and inventory records use database transactions and referential integrity.
- Eligibility, shelf-life and notification thresholds remain configurable because approved rules may change.
- The interface uses forms and tables suitable for a semester-scale implementation.

2.6 Assumptions and Dependencies

- Each hospital and staff account is verified by an administrator before activation.
- Laboratory results are entered by authorized staff; direct analyser integration is outside the first release.
- The deployment provides working email, database and backup services.
- A qualified authority approves operational eligibility and retention rules before deployment.

2.7 Items Outside Scope

- Payment processing and insurance claims.
- Direct control of laboratory analysers or storage hardware.
- Biometric or national identity integration.
- Automated clinical diagnosis or replacement of professional cross-match review.

3 External Interface Requirements

3.1 User Interfaces

Role-specific dashboards present only permitted actions. Forms display mandatory-field markers, inline validation and a summary of submission errors. Tables support paging, filtering and clear empty states. Destructive actions require confirmation. Status values use text labels and do not rely on colour alone.

3.2 Hardware Interfaces

No dedicated hardware interface is required for the first release. Staff use standard desktop or tablet input devices. Barcode support may be added later through keyboard-compatible scanners.

3.3 Software Interfaces

Interface	Purpose	Failure handling
PostgreSQL	Persistent transactional storage	Rollback transaction and log a database error
SMTP email provider	Appointments and operational notifications	Record failure and retry through a scheduled task
Backup storage	Encrypted database backup retention	Alert administrator and preserve prior successful backups

3.4 Communication Interfaces

Browser traffic uses HTTPS. API payloads use JSON encoded as UTF-8. Server-generated timestamps use ISO 8601 UTC format. Email transport uses the configured authenticated and encrypted provider connection.

4 System Requirements

4.1 Requirement Quality Rules

Every requirement has one identifier, one primary behaviour and one verification reference. Quantitative limits are included where acceptance depends on time, load, retention or viewport size. Configurable values are identified instead of being embedded in application code.

4.2 Functional Requirements

Table 3 Functional requirements

ID	Requirement	Priority	Verification
FR-01	The system shall register a donor after validating mandatory identity, contact, consent and medical history fields.	High	Inspection and TC-F01
FR-02	The system shall maintain one donor profile per registered email address and shall reject a duplicate active profile.	Medium	TC-F01
FR-03	The system shall evaluate donor eligibility using administrator-configured rules for age, weight, donation type, donation interval, deferrals and screening responses.	High	TC-F02
FR-04	The system shall display each failed eligibility rule and shall prevent appointment confirmation when any blocking rule fails.	High	TC-F02
FR-05	The system shall allow an eligible donor to create, reschedule or cancel an appointment for an available branch and time slot.	High	TC-F03
FR-06	The system shall prevent appointment capacity from exceeding the configured capacity of a time slot.	High	TC-F03
FR-07	The system shall display the authenticated donor's donation history, appointment history and eligibility date.	Medium	TC-F04
FR-08	The system shall allow authorized staff to record temporary or permanent donor deferrals with reason, start date, end date when applicable and approving staff member.	High	TC-F05
FR-09	The system shall assign a unique donation identifier and unique blood unit identifier to every recorded donation and derived component.	High	TC-F06
FR-10	The system shall record collection time, volume, blood group, component type, collector and source donor for each blood unit.	High	TC-F06
FR-11	The system shall store laboratory results for the configured mandatory transfusion-transmitted infection test panel for each unit.	High	TC-F07
FR-12	The system shall prevent a unit from entering the Available state until all mandatory test results are recorded as acceptable by authorized staff.	High	TC-F07
FR-13	The system shall place a unit in Quarantined state when any mandatory screening result is reactive, indeterminate or incomplete.	High	TC-F07

ID	Requirement	Priority	Verification
FR-14	The system shall manage unit states as Collected, Testing, Available, Reserved, Issued, Quarantined, Expired or Discarded and shall reject an invalid state transition.	High	TC-F08
FR-15	The system shall calculate expiry from collection time, component type and the configured shelf-life rule.	High	TC-F08
FR-16	The system shall mark an unissued unit Expired when its expiry time is reached and shall exclude it from available stock.	High	TC-F08
FR-17	The system shall allow authorized users to search available stock by blood group, component, branch and expiry date.	High	TC-F09
FR-18	The system shall notify designated staff when available stock falls below a configured threshold or when a unit enters the configured near-expiry period.	Medium	TC-F10
FR-19	The system shall allow hospital staff to create a request containing blood group, component, quantity, priority, required-by time and clinical reference.	High	TC-F11
FR-20	The system shall support Pending, Approved, Partially Fulfilled, Fulfilled, Rejected and Cancelled request states and shall record the actor and time of every transition.	High	TC-F11
FR-21	The system shall allow hospital staff to cancel only a Pending request and shall require blood bank staff to cancel a request after approval.	Medium	TC-F12
FR-22	The system shall allow authorized staff to reserve compatible Available units for an approved request without assigning one unit to more than one active request.	High	TC-F13
FR-23	The system shall record cross-match outcome before a reserved unit is issued.	High	TC-F13
FR-24	The system shall complete unit issuance and request status update in one database transaction.	High	TC-F13
FR-25	The system shall authenticate Donor, Blood Bank Staff, Hospital Staff and System Administrator accounts before protected functions are accessed.	High	TC-S01
FR-26	The system shall allow a system administrator to create, deactivate and assign one approved role to a user account.	High	TC-S02
FR-27	The system shall allow a system administrator to manage branches, time-slot capacities, eligibility rules, shelf-life rules and stock thresholds.	Medium	TC-F14

ID	Requirement	Priority	Verification
FR-28	The system shall generate inventory, expiry, donation, request and audit reports for a selected date range and branch.	Medium	TC-F15
FR-29	The system shall send appointment reminders and request-status notifications through the configured email service and shall record delivery success or failure.	Low	TC-F16
FR-30	The system shall write an audit event for authentication, deferral, screening, inventory, request, issuance, configuration and account-management changes.	High	TC-S03

Table 4 Non-functional requirements

ID	Requirement	Quality area	Verification
NFR-01	For at least 95 percent of search and dashboard requests, the server shall return a response within 3 seconds with 100 concurrent authenticated users in the defined test environment.	Performance	TC-N01
NFR-02	The system shall complete a stock reservation transaction within 5 seconds under the same load and shall prevent duplicate reservation under concurrent requests.	Performance	TC-N02
NFR-03	The monthly service availability target shall be 99.0 percent, excluding planned maintenance announced at least 24 hours in advance.	Reliability	Operational review
NFR-04	A scheduled encrypted backup shall run every 24 hours, retain seven daily copies and four weekly copies, and support restoration within 4 hours during the acceptance test.	Recovery	TC-N03
NFR-05	The system shall preserve committed donation, screening, reservation and issuance data after an application restart.	Reliability	TC-N04
NFR-06	The interface shall support the latest two major versions of Chrome, Edge and Firefox and viewport widths from 360 to 1440 pixels without horizontal scrolling on standard forms.	Compatibility	TC-N05
NFR-07	A first-time staff user shall complete donor lookup and appointment check-in within 5 minutes using the supplied task instructions and without developer assistance.	Usability	TC-N06
NFR-08	The application shall separate presentation, business rules and data-access code into documented modules with automated unit tests for eligibility, unit state transitions and request state transitions.	Maintainability	Code review
NFR-09	The database schema shall support multiple branches while enforcing branch-level access restrictions for operational users.	Scalability	TC-S02
NFR-10	The application shall record dates and times in UTC and display them in the branch time zone configured by the administrator.	Data integrity	TC-N07
NFR-11	The system shall not expose medical-history or screening-result fields in availability search results or hospital-facing reports.	Privacy	TC-S04
NFR-12	The system shall support retention and export rules configured to match the approved institutional and regulatory record-retention policy.	Compliance	Inspection

5 Data Requirements

5.1 Core Data Entities

The core entities are User, Donor, Branch, Appointment, Deferral, Donation, BloodUnit, LabResult, BloodRequest, Reservation, CrossMatch, Issue, Notification and AuditEvent. Every clinical and inventory record has a stable identifier and creation timestamp.

5.2 Data Integrity Rules

- A blood unit identifier is unique and cannot be reused.
- A lab result references one valid blood unit and one configured test type.
- An issued unit references one issue transaction and cannot return to Available state.
- A reserved unit belongs to at most one active request.
- Deletion of clinical, inventory and audit records is disabled through normal application functions.

5.3 Retention and Privacy

Retention periods are configuration values approved by the institution. Export and archival processes preserve identifiers, timestamps and audit references. Availability searches expose stock counts and unit suitability fields but do not expose donor medical history or laboratory details to hospital accounts.

6 Security Objectives and Requirements

6.1 Security Objectives

1. Protect the confidentiality of donor, medical and request information.
2. Preserve the integrity of screening, inventory, cross-match and issuance records.
3. Permit each role to perform only authorized actions within its assigned scope.
4. Maintain accountable and reviewable records of security-sensitive and operational changes.
5. Support recovery from data loss without exposing backup contents.

6.2 Security Requirements

Table 5 Security requirements

ID	Requirement	Verification
SEC-01	The system shall enforce server-side authorization for every protected action according to the role-permission matrix.	TC-S02
SEC-02	The system shall store passwords using Django's approved adaptive password hashing mechanism and shall never store or log a plain-text password.	Code review
SEC-03	The system shall lock an account for 15 minutes after five consecutive failed login attempts within 10 minutes and shall record the event.	TC-S01
SEC-04	The system shall expire an authenticated session after 15 minutes of inactivity and shall invalidate the session after logout or account deactivation.	TC-S05
SEC-05	The production deployment shall redirect HTTP to HTTPS and shall transmit authenticated traffic only over TLS.	Configuration inspection
SEC-06	The system shall protect state-changing requests against cross-site request forgery and shall validate and encode user-controlled input before storage or display.	TC-S06
SEC-07	Audit records shall include actor ID, event type, UTC timestamp, affected record ID and outcome and shall not be editable through normal application roles.	TC-S03
SEC-08	Database backups shall be encrypted and accessible only to the backup service account and designated administrator.	TC-N03
SEC-09	The system shall display only the minimum donor and patient information required for the current role and task.	TC-S04
SEC-10	The system shall return a generic authentication failure message that does not reveal whether an email address is registered.	TC-S01

6.3 Role Permission Matrix

Capability	Donor	Blood Bank Staff	Hospital Staff	Administrator
Own profile and appointments	Create and update	View for operations	No access	Account status only
Donor medical history	Own record	Authorized view and update	No access	No routine access
Laboratory results	No access	Create and view	No access	No routine access
Inventory	No access	Full branch operations	Restricted availability	Reports and configuration
Blood requests	No access	Approve and fulfil	Create and track own hospital	Reports only
Accounts and configuration	No access	No access	No access	Create and manage
Audit records	No access	Restricted read	No access	Read and export

7 Use Case Model

7.1 Actors

The primary actors are Donor, Blood Bank Staff, Hospital Staff and System Administrator. Email and scheduled-task services support system behaviour but are not human actors.

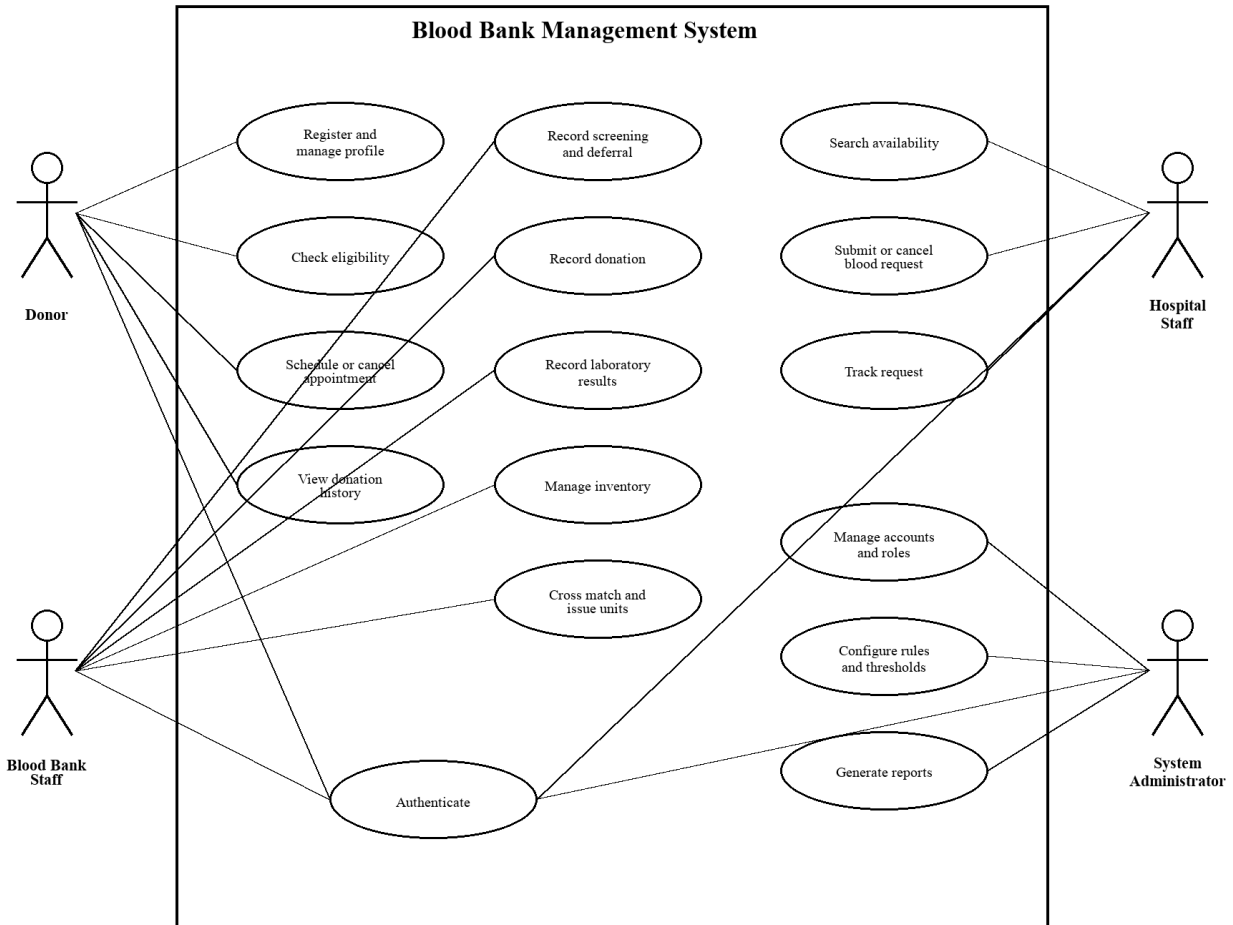


Figure 1 UML use case diagram

7.2 Detailed Use Cases

UC-01 Check Donor Eligibility

Field	Description
Primary actor	Donor or Blood Bank Staff
Preconditions	An identified donor profile exists.
Trigger	An eligibility check is requested.
Main flow	The system loads current profile data, donation history, active deferrals and configured rules. It evaluates each rule and returns Eligible or Not Eligible with reasons.
Alternate and exception flow	Missing mandatory data produces an Incomplete decision. Rule-service failure produces no eligibility approval.
Postconditions	The decision and evaluation time are recorded.

UC-02 Record Donation and Units

Field	Description
Primary actor	Blood Bank Staff
Preconditions	The donor is checked in and approved for donation.
Trigger	Staff record a completed collection.
Main flow	The system validates collection data, creates one donation identifier, creates unique component unit identifiers and sets each unit to Collected.
Alternate and exception flow	Duplicate identifiers or invalid volume values are rejected.
Postconditions	Donation and unit records exist with an audit event.

UC-03 Record Screening Results

Field	Description
Primary actor	Blood Bank Staff
Preconditions	A unit exists in Testing state.
Trigger	Staff submit laboratory results.
Main flow	The system validates the test panel. An acceptable complete panel permits release to Available. A reactive, indeterminate or incomplete panel produces Quarantined status.
Alternate and exception flow	Unauthorized access, missing test type or invalid result value is rejected.
Postconditions	The unit status and audit record are updated.

UC-04 Submit Blood Request

Field	Description
Primary actor	Hospital Staff
Preconditions	The hospital account is active.
Trigger	Hospital staff submit required blood details.
Main flow	The system validates group, component, quantity, priority, required-by time and clinical reference, then creates a Pending request.
Alternate and exception flow	Missing data is rejected. Duplicate client submission returns the existing request when the idempotency key matches.
Postconditions	A traceable Pending request exists.

UC-05 Reserve and Issue Blood Units

Field	Description
Primary actor	Blood Bank Staff
Preconditions	An Approved request and compatible Available units exist.
Trigger	Staff select units for fulfilment.
Main flow	The system atomically reserves units, records cross-match results and issues acceptable units. The request becomes Partially Fulfilled or Fulfilled.
Alternate and exception flow	Failed cross-match releases the unit according to policy. Concurrent reservation conflict rejects the later request.
Postconditions	Unit and request states agree and an audit event exists.

UC-06 Manage Accounts and Roles

Field	Description
Primary actor	System Administrator
Preconditions	The administrator is authenticated.
Trigger	An account is created, changed or deactivated.
Main flow	The system validates the role, branch scope and account status before saving the change.
Alternate and exception flow	The final active administrator cannot be deactivated through the interface.
Postconditions	The account change and audit event are stored.

8 Requirements Traceability

Table 7 Requirements traceability matrix

Requirement	Design allocation	Verification
FR-01	Donor Service	Inspection and TC-F01
FR-02	Donor Service	TC-F01
FR-03	Donor Service	TC-F02
FR-04	Donor Service	TC-F02
FR-05	Donor Service	TC-F03
FR-06	Donor Service	TC-F03
FR-07	Donor Service	TC-F04
FR-08	Donor Service	TC-F05
FR-09	Donation and Screening Service	TC-F06
FR-10	Donation and Screening Service	TC-F06
FR-11	Donation and Screening Service	TC-F07
FR-12	Donation and Screening Service	TC-F07
FR-13	Donation and Screening Service	TC-F07
FR-14	Donation and Screening Service	TC-F08
FR-15	Donation and Screening Service	TC-F08
FR-16	Donation and Screening Service	TC-F08
FR-17	Inventory and Request Service	TC-F09
FR-18	Inventory and Request Service	TC-F10
FR-19	Inventory and Request Service	TC-F11
FR-20	Inventory and Request Service	TC-F11
FR-21	Inventory and Request Service	TC-F12
FR-22	Inventory and Request Service	TC-F13
FR-23	Inventory and Request Service	TC-F13
FR-24	Inventory and Request Service	TC-F13
FR-25	Authentication, Administration or Reporting	TC-S01
FR-26	Authentication, Administration or Reporting	TC-S02
FR-27	Authentication, Administration or Reporting	TC-F14
FR-28	Authentication, Administration or Reporting	TC-F15
FR-29	Authentication, Administration or Reporting	TC-F16
FR-30	Authentication, Administration or Reporting	TC-S03
NFR-01	Cross-cutting architecture	TC-N01
NFR-02	Cross-cutting architecture	TC-N02
NFR-03	Cross-cutting architecture	Operational review
NFR-04	Cross-cutting architecture	TC-N03
NFR-05	Cross-cutting architecture	TC-N04
NFR-06	Cross-cutting architecture	TC-N05

Requirement	Design allocation	Verification
NFR-07	Cross-cutting architecture	TC-N06
NFR-08	Cross-cutting architecture	Code review
NFR-09	Cross-cutting architecture	TC-S02
NFR-10	Cross-cutting architecture	TC-N07
NFR-11	Cross-cutting architecture	TC-S04
NFR-12	Cross-cutting architecture	Inspection
SEC-01	Security architecture	TC-S02
SEC-02	Security architecture	Code review
SEC-03	Security architecture	TC-S01
SEC-04	Security architecture	TC-S05
SEC-05	Security architecture	Configuration inspection
SEC-06	Security architecture	TC-S06
SEC-07	Security architecture	TC-S03
SEC-08	Security architecture	TC-N03
SEC-09	Security architecture	TC-S04
SEC-10	Security architecture	TC-S01

Part B Software Test Plan

Document identifier BBMS STP 1.0

1 Introduction

1.1 Purpose

This Test Plan defines the test scope, approach, environment, responsibilities, acceptance criteria and traceability for the Blood Bank Management System. It covers functional, integration, performance, compatibility, recovery and security validation.

1.2 Test Objectives

- Verify that every high-priority workflow satisfies its linked SRS requirements.
- Verify that invalid input, authorization failures and concurrent operations do not corrupt data.
- Measure the specified performance, compatibility and recovery requirements.
- Provide repeatable evidence for acceptance of the academic prototype.

1.3 References

This plan uses the Part A requirement baseline and the standards listed in the final References section.

1.4 Test Items

- Django web application and REST-style endpoints.
- PostgreSQL schema, constraints and transactional behaviour.
- Role-specific browser interfaces.
- Scheduled alert, expiry and backup tasks.
- Email-provider integration through a controlled test service.

2 Test Scope

2.1 Included

Testing includes all functional requirements, security requirements and measurable non-functional requirements that can be evaluated in the academic environment. Unit, integration, system and acceptance-level checks are included.

2.2 Excluded

The plan excludes clinical certification, laboratory-analyser integration, biometric identity verification, national-scale load testing and assessment of physical blood storage conditions. These exclusions do not remove the need for controlled validation before real deployment.

3 Features to Be Tested

Feature	Requirement groups	Primary test level
Donor and appointment management	FR-01 to FR-08	System and acceptance
Donation, screening and unit lifecycle	FR-09 to FR-16	Unit, integration and system
Inventory, requests and issue	FR-17 to FR-24	Integration, system and concurrency
Authentication, administration, reports and audit	FR-25 to FR-30	Security and system
Quality requirements	NFR-01 to NFR-12	Performance, recovery, compatibility and review
Security controls	SEC-01 to SEC-10	Security test and code review

3.1 Features Not to Be Tested

Third-party email infrastructure availability and operating-system security are outside direct product testing. Integration tests use a controlled email service. Infrastructure configuration is inspected separately.

4 Test Approach

4.1 Test Levels

Level	Focus	Examples
Unit	Business-rule and state-transition functions	Eligibility, expiry and request transitions
Integration	Database, email and scheduled-task boundaries	Unit release, alerts and transaction rollback
System	End-to-end role workflows	Registration, screening and issue
Acceptance	SRS pass criteria in a representative environment	Performance, recovery and supported browsers

4.2 Test Techniques

- Equivalence partitioning for valid and invalid form input.
- Boundary-value analysis for age, weight, interval, capacity, quantity and time limits.
- State-transition testing for units and blood requests.
- Concurrency testing for appointment capacity and unit reservation.
- Negative authorization tests using direct URLs and API requests.
- Performance tests with measured response percentiles.
- Backup restoration and browser compatibility checks.

4.3 Test Environment

Area	Planned environment
Application	Django application configured with production-like settings except controlled test credentials
Database	PostgreSQL with representative test data and isolated test schema
Browsers	Latest two major versions of Chrome, Edge and Firefox available during acceptance
Email	Local or sandbox SMTP service that records sent messages
Load generation	Scripted HTTP clients producing 100 authenticated concurrent users
Security	Automated dependency and static checks plus manual authorization tests

5 Test Management

5.1 Security Validation

Security validation covers authentication failure handling, account lockout, session expiry, role and branch authorization, CSRF protection, input handling, medical-data exposure, audit completeness and backup access. Tests use synthetic records only. No real donor or patient data is used.

Security area	Validation method	Evidence
Authentication	Failed-login, lockout, logout and session-expiry tests	Request log and audit event
Authorization	Role matrix, direct URL and cross-branch access tests	HTTP result and audit event
Input protection	CSRF and stored-script payload tests	Rejected request or safely encoded display
Audit	Required-field and edit-prevention tests	Exported audit records
Privacy	Hospital and report field inspection	Screenshots and response-body comparison
Backup	Encryption, access restriction and restoration test	Backup log and restored record comparison

5.2 Entry Criteria

- The requirement baseline and database schema are versioned.
- The selected build installs successfully in the test environment.
- Required test accounts and synthetic data are available.
- Critical unit tests pass before system testing begins.

5.3 Exit Criteria

- All listed test cases have an execution result and evidence.
- All high-priority functional and security tests pass.
- No open defect causes data corruption, unauthorized access or incorrect issue of a unit.

- At least 90 percent of all planned test cases pass, with remaining defects documented and accepted by the evaluator.
- Performance and backup-restoration acceptance criteria pass in the defined environment.

5.4 Suspension and Resumption Criteria

Testing is suspended when the build cannot start, the test database is unavailable, a defect corrupts shared test data or more than 20 percent of attempted cases are blocked by one defect. Testing resumes after the cause is fixed, the environment is restored and a smoke test passes.

5.5 Pass and Fail Rules

A test passes only when the observed result matches every stated expected result. A difference in stored data, authorization decision, state transition, response-time limit or displayed result causes failure. Blocked tests are not counted as passed.

5.6 Deliverables

- Approved Test Plan
- Test data set and scripts
- Execution log and evidence
- Defect log
- Test summary report

5.7 Responsibilities

Role	Responsibility
Test coordinator	Maintain plan, environment, execution order and summary
Tester	Prepare data, execute cases, save evidence and report defects
Developer	Support diagnosis, correct defects and provide a new build
Team reviewer	Check traceability and confirm exit criteria

5.8 Risks and Mitigation

Risk	Effect	Mitigation
Limited project time	Reduced regression coverage	Automate unit tests and run high-priority cases first
Unstable external email service	Blocked notification tests	Use a local or sandbox SMTP service
Insufficient representative data	Unrealistic performance result	Generate synthetic data at the stated scale
Shared test records	Tests influence one another	Reset an isolated database before each suite

6 Test Cases

The following nineteen cases exceed the required minimum of seven to ten and cover functional, non-functional and security requirements.

6.1 TC-F01 Donor registration and duplicate prevention

Field	Specification
Requirement links	FR-01, FR-02
Test type	Functional
Preconditions	No active donor exists for donor1@example.test.
Test data	Valid mandatory fields, consent accepted, email donor1@example.test.
Procedure	1. Submit a complete donor registration. 2. Submit the same email address again.
Expected result and pass criterion	The first request creates one donor profile. The second request is rejected with a duplicate-account message and no second profile is stored.

6.2 TC-F02 Eligibility rules and failed-reason display

Field	Specification
Requirement links	FR-03, FR-04
Test type	Functional and boundary
Preconditions	Eligibility rules are configured according to the approved policy.
Test data	Eligible donor, under-age donor, active-deferral donor, and donor inside the configured donation interval.
Procedure	Evaluate eligibility for each test profile and attempt appointment confirmation.
Expected result and pass criterion	Only the eligible profile can confirm an appointment. Each blocked profile receives the applicable reason without disclosure of unrelated medical data.

6.3 TC-F03 Appointment capacity and cancellation

Field	Specification
Requirement links	FR-05, FR-06
Test type	Functional and concurrency
Preconditions	A slot with capacity one exists and two eligible donors are active.
Test data	Two concurrent appointment requests for the same slot.
Procedure	Submit both requests concurrently. Cancel the successful booking and retry the rejected request.
Expected result and pass criterion	Only one initial booking succeeds. After cancellation, the second donor can book the released capacity.

6.4 TC-F04 Donor history access

Field	Specification
Requirement links	FR-07, SEC-01
Test type	Functional and security
Preconditions	Two donors with different records exist.
Test data	Authenticated donor A and record identifiers for donors A and B.
Procedure	Open donor A history. Attempt to request donor B history.
Expected result and pass criterion	Donor A can view only donor A records. Access to donor B records is denied and logged.

6.5 TC-F05 Donor deferral

Field	Specification
Requirement links	FR-08
Test type	Functional
Preconditions	Authorized staff and active donor exist.
Test data	Temporary deferral with valid reason and end date, followed by a permanent deferral without an end date.
Procedure	Record each valid deferral and attempt an invalid temporary deferral without an end date.
Expected result and pass criterion	Valid deferrals are stored with staff identity. The incomplete temporary deferral is rejected. Eligibility reflects the active deferral.

6.6 TC-F06 Donation and unique unit creation

Field	Specification
Requirement links	FR-09, FR-10
Test type	Functional and data integrity
Preconditions	Eligible checked-in donor and authorized staff exist.
Test data	One whole-blood donation with component preparation instructions.
Procedure	Record the collection and generate component units. Attempt to reuse a generated unit ID.
Expected result and pass criterion	The donation and units are stored with unique IDs and complete collection metadata. Duplicate unit ID storage fails.

6.7 TC-F07 Laboratory screening and quarantine

Field	Specification
Requirement links	FR-11, FR-12, FR-13
Test type	Functional and negative
Preconditions	Collected unit exists in Testing state.
Test data	Complete acceptable panel, incomplete panel, and panel containing one reactive result.
Procedure	Submit each result set for separate units and request release to inventory.
Expected result and pass criterion	The acceptable unit becomes Available. Incomplete and reactive units remain unavailable and enter Quarantined state as configured.

6.8 TC-F08 Unit state transitions and expiry

Field	Specification
Requirement links	FR-14, FR-15, FR-16
Test type	Functional and scheduled job
Preconditions	Shelf-life rules exist for the tested component.
Test data	Available unit just before expiry, after expiry, and an invalid direct Collected-to-Issued transition.
Procedure	Run the expiry task and attempt the invalid state transition.
Expected result and pass criterion	The expired unit becomes Expired and disappears from available stock. The invalid transition is rejected and logged.

6.9 TC-F09 Authorized inventory search

Field	Specification
Requirement links	FR-17, NFR-11
Test type	Functional and privacy
Preconditions	Available and unavailable units exist across branches.
Test data	Blood group, component, branch and expiry filters.
Procedure	Search as staff and hospital staff with the same filters.
Expected result and pass criterion	Both roles see permitted availability data. Medical history and screening details are absent. Unavailable units are excluded.

6.10 TC-F10 Low-stock and near-expiry alerts

Field	Specification
Requirement links	FR-18
Test type	Functional and integration
Preconditions	Thresholds, recipients and near-expiry period are configured.
Test data	Stock count crossing below threshold and unit entering near-expiry period.
Procedure	Run inventory update and scheduled alert task twice.
Expected result and pass criterion	One alert per new condition is sent and recorded. The repeated run does not create a duplicate alert for the unchanged condition.

6.11 TC-F11 Blood request workflow

Field	Specification
Requirement links	FR-19, FR-20
Test type	Functional and workflow
Preconditions	Hospital and staff accounts are active.
Test data	Valid request and invalid request without required-by time.
Procedure	Submit both requests. Approve the valid request and reject another valid request.
Expected result and pass criterion	The incomplete request is rejected. Valid state changes store actor and UTC time and follow the permitted transition rules.

6.12 TC-F12 Request cancellation permissions

Field	Specification
Requirement links	FR-21, SEC-01
Test type	Functional and security
Preconditions	One Pending and one Approved request exist.
Test data	Authenticated hospital staff and blood bank staff accounts.
Procedure	Hospital staff cancel both requests. Blood bank staff cancel the approved request.
Expected result and pass criterion	Hospital staff can cancel only the Pending request. Staff can cancel the Approved request. All attempts are logged.

6.13 TC-F13 Reservation, cross-match and atomic issue

Field	Specification
Requirement links	FR-22, FR-23, FR-24
Test type	Functional, concurrency and integrity
Preconditions	Approved requests and compatible Available units exist.
Test data	Two concurrent reservations for the same unit, failed cross-match, and successful issue.
Procedure	Run concurrent reservations. Record failed cross-match. Record successful cross-match and issue. Simulate failure during issue commit.
Expected result and pass criterion	One reservation succeeds. Failed cross-match prevents issue. Successful issue updates unit and request atomically. Simulated failure rolls back both updates.

6.14 TC-S01 Login failure, lockout and generic message

Field	Specification
Requirement links	FR-25, SEC-03, SEC-10
Test type	Security
Preconditions	One active account exists.
Test data	Incorrect passwords for registered and unregistered emails.
Procedure	Submit five failed attempts within 10 minutes and then submit correct credentials during the lockout period.
Expected result and pass criterion	Both email cases show the same generic failure message. The account locks for 15 minutes, correct credentials are rejected during lockout, and the event is audited.

6.15 TC-S02 Role-based authorization

Field	Specification
Requirement links	FR-26, SEC-01, NFR-09
Test type	Security
Preconditions	Accounts for all four roles exist.
Test data	Representative URLs and actions for donor, staff, hospital staff and administrator.
Procedure	Attempt each protected action using every role, including a request for another branch outside the assigned scope.
Expected result and pass criterion	Only permissions in the approved role matrix succeed. Direct URL access and out-of-scope branch access are denied and logged.

6.16 TC-S03 Audit completeness and immutability

Field	Specification
Requirement links	FR-30, SEC-07
Test type	Security and audit
Preconditions	Authorized operational and administrator accounts exist.
Test data	Screening edit, issue, role change and rejected authorization attempt.
Procedure	Perform each event, inspect the audit record, and attempt to edit an audit event through the application.
Expected result and pass criterion	Every event records actor, type, UTC time, record ID and outcome. No normal role can edit or delete an audit event.

6.17 TC-N01 Search and dashboard response time

Field	Specification
Requirement links	NFR-01
Test type	Performance
Preconditions	Representative database contains at least 10,000 units and 5,000 donors.
Test data	100 concurrent authenticated users for 15 minutes.
Procedure	Run representative search and dashboard requests and record server response times.
Expected result and pass criterion	At least 95 percent of measured responses complete within 3 seconds and no request returns an integrity error.

6.18 TC-N03 Encrypted backup and restoration

Field	Specification
Requirement links	NFR-04, SEC-08
Test type	Recovery and security
Preconditions	Backup schedule and restricted storage are configured.
Test data	Known donor, unit, request and audit records.
Procedure	Run backup, verify encryption and access restriction, restore into an empty acceptance environment, and compare known records.
Expected result and pass criterion	The authorized service creates and reads the encrypted backup. Unauthorized access fails. Restoration completes within 4 hours with all known records intact.

6.19 TC-N05 Responsive browser layout

Field	Specification
Requirement links	NFR-06
Test type	Compatibility and usability
Preconditions	Supported browser versions are available.
Test data	Chrome, Edge and Firefox at 360, 768 and 1440 pixel viewport widths.
Procedure	Open registration, appointment, inventory and request forms at each size and complete one valid submission.
Expected result and pass criterion	No standard form requires horizontal scrolling, labels remain associated with controls, and each submission completes successfully.

7 Test Traceability

Table 16 Test-to-requirement traceability

Test case	Requirement links	Coverage	Initial status
TC-F01	FR-01, FR-02	Functional	Not executed
TC-F02	FR-03, FR-04	Functional and boundary	Not executed
TC-F03	FR-05, FR-06	Functional and concurrency	Not executed
TC-F04	FR-07, SEC-01	Functional and security	Not executed
TC-F05	FR-08	Functional	Not executed
TC-F06	FR-09, FR-10	Functional and data integrity	Not executed
TC-F07	FR-11, FR-12, FR-13	Functional and negative	Not executed
TC-F08	FR-14, FR-15, FR-16	Functional and scheduled job	Not executed
TC-F09	FR-17, NFR-11	Functional and privacy	Not executed
TC-F10	FR-18	Functional and integration	Not executed
TC-F11	FR-19, FR-20	Functional and workflow	Not executed
TC-F12	FR-21, SEC-01	Functional and security	Not executed
TC-F13	FR-22, FR-23, FR-24	Functional, concurrency and integrity	Not executed
TC-S01	FR-25, SEC-03, SEC-10	Security	Not executed
TC-S02	FR-26, SEC-01, NFR-09	Security	Not executed
TC-S03	FR-30, SEC-07	Security and audit	Not executed
TC-N01	NFR-01	Performance	Not executed
TC-N03	NFR-04, SEC-08	Recovery and security	Not executed
TC-N05	NFR-06	Compatibility and usability	Not executed

8 Test Schedule

Activity	Planned sequence	Output
Unit and integration preparation	1	Automated checks and test data
Functional system testing	2	Functional execution results
Security and recovery testing	3	Security and restoration evidence
Performance and compatibility testing	4	Measurements and browser evidence
Regression and summary	5	Final test summary and defect status

Part C Software Architecture and Design Specification

Document identifier BBMS SADS 1.0

1 Introduction

1.1 Purpose

This specification describes the architecture, components, data model, security controls, interactions, APIs and error-handling design used to satisfy the Part A requirements.

1.2 Design Scope

The design covers a Django and PostgreSQL web application deployed behind Nginx. It focuses on a semester-scale system while preserving clear module boundaries and branch-aware data for later growth.

1.3 Design Principles

- Keep business rules outside browser templates and database views.
- Enforce authorization and validation on the server.
- Use transactions for reservation, issue and other multi-record state changes.
- Keep eligibility and shelf-life rules configurable and auditable.
- Expose the minimum information required for each role.

2 Architectural Drivers

Driver	Design response
Sensitive donor and medical data	RBAC, branch scope, HTTPS, minimal disclosure and audit logging
Inventory integrity	State machine, database constraints and transactional reservation and issue
Changing operational rules	Configuration tables and rule service
Semester implementation scope	Modular monolith rather than distributed services
Future branch expansion	Branch identifiers and scoped queries in the initial schema
Testability	Service-layer functions and explicit component interfaces

3 Architecture Description

3.1 Selected Pattern

The system uses a layered modular monolith. Browser-facing views call service-layer modules, service modules apply business rules and data-access code uses the Django object-relational mapper. This pattern is appropriate for a small team because deployment remains simple while domain boundaries remain testable.

3.2 Layer Responsibilities

Layer	Responsibilities
Presentation	Django views, templates, forms, API serializers and role-specific navigation
Application services	Use-case orchestration, authorization checks and transaction boundaries
Domain services	Eligibility, unit lifecycle, compatibility, request workflow and alert rules
Data access	Django models, query services, constraints and migrations
Infrastructure	Email, scheduled tasks, logs, backups, Nginx and database connections

3.3 Component Diagram

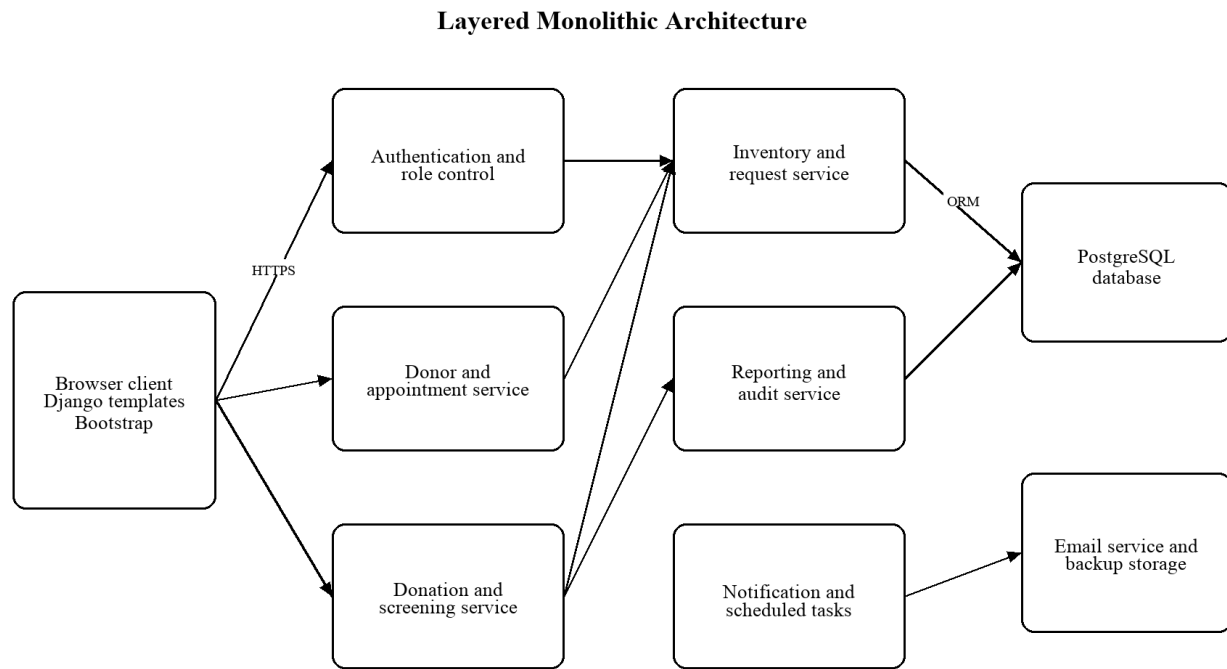


Figure 2 Component diagram

3.4 Deployment View

A browser connects to Nginx over HTTPS. Nginx forwards application traffic to Gunicorn workers running Django. Django uses a restricted database account to connect to PostgreSQL. Scheduled tasks share application code but run under a separate process. Email and backup storage use distinct restricted credentials.

4 Component Design

Component	Main responsibilities	Requirement allocation
Authentication and Role Control	Login, logout, sessions, account status and authorization	FR-25, FR-26, SEC-01 to SEC-05, SEC-10
Donor and Appointment Service	Profiles, eligibility, deferrals, slots and donor history	FR-01 to FR-08
Donation and Screening Service	Collection, components, test panel and quarantine	FR-09 to FR-13
Inventory and Request Service	Unit states, expiry, search, requests, reservation, cross-match and issue	FR-14 to FR-24
Administration Service	Branches, configurable rules and thresholds	FR-27
Reporting and Audit Service	Reports and append-only application audit events	FR-28, FR-30, SEC-07
Notification and Task Service	Email, reminders, alerts, expiry task and backup trigger	FR-18, FR-29, NFR-04

4.1 Unit State Machine

Current state	Permitted next states	Condition
Collected	Testing or Discarded	Testing begins or collection is rejected
Testing	Available, Quarantined or Discarded	Mandatory panel and review determine outcome
Available	Reserved, Expired or Discarded	Request allocation, time expiry or authorized disposal
Reserved	Available, Issued or Expired	Reservation release, successful issue or expiry
Issued	None	Terminal state
Quarantined	Discarded	Authorized disposal
Expired	Discarded	Authorized disposal
Discarded	None	Terminal state

4.2 Request State Machine

Current state	Permitted next states
Pending	Approved, Rejected or Cancelled
Approved	Partially Fulfilled, Fulfilled or Cancelled
Partially Fulfilled	Fulfilled or Cancelled
Fulfilled	None
Rejected	None
Cancelled	None

5 Data Design

Logical Data Model

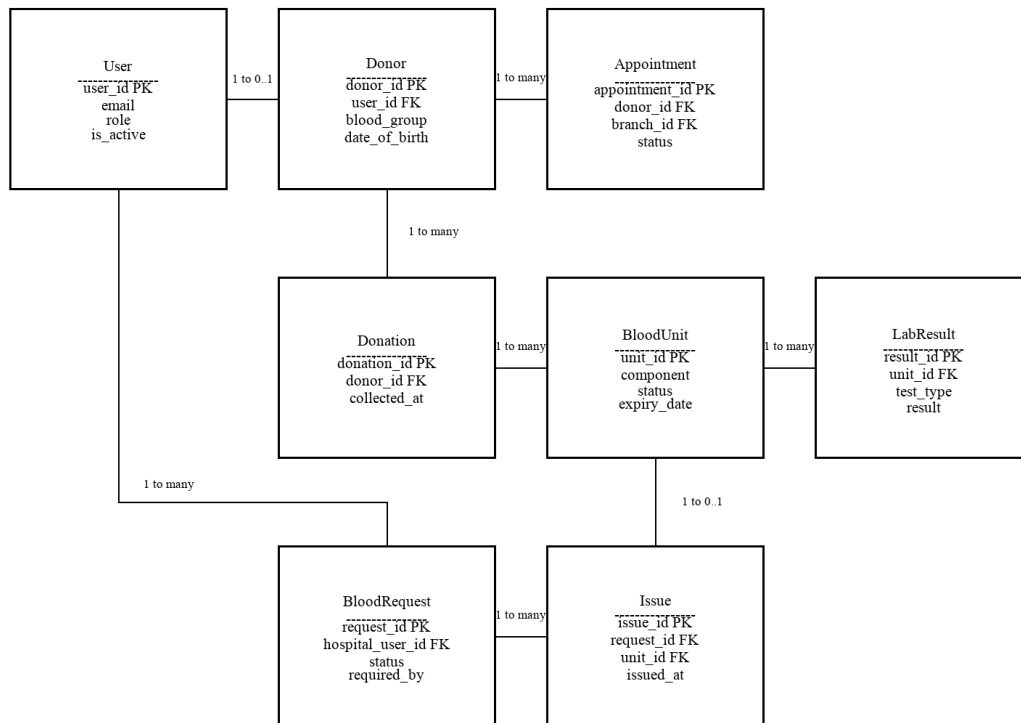


Figure 3 Logical data model

5.1 Database Constraints

- Unique constraints protect account email, donation ID and unit ID.
- Foreign keys prevent orphaned donation, result, request and issue records.
- Database checks restrict enumerated state values and positive quantities.
- A conditional unique constraint prevents more than one active reservation for a unit.
- Transactional service methods lock candidate unit rows during reservation and issue.

5.2 Audit Design

AuditEvent stores actor, branch, event type, affected record type, affected record ID, UTC timestamp, outcome and a structured summary of changed fields. Application roles have no update or delete path for this table. Database and backup access remain restricted to service accounts and designated administrators.

6 Security Architecture

6.1 Trust Boundaries

- The public network ends at Nginx, which terminates HTTPS and forwards only expected traffic.
- The Django application is the only normal path to the database.
- Email and backup services use separate credentials and receive only required data.
- Browser input is untrusted and is validated at form, serializer and service boundaries.

6.2 Authentication and Session Design

Django authentication provides password hashing and session management. The login service applies generic errors and account lockout. Session middleware applies inactivity timeout. Deactivation invalidates active sessions through a recorded account-status change.

6.3 Authorization Design

Each application service checks the account role, active status and branch scope before reading or changing protected records. Templates hide unauthorized actions, but server checks remain authoritative. Object-level filters prevent access by guessed identifiers.

6.4 Input and Output Protection

Django forms and serializers validate type, length, range and permitted values. CSRF tokens protect browser state changes. Templates use automatic output escaping. File upload is outside the initial release, which reduces the attack surface.

6.5 Logging Backup and Recovery

Security events and operational changes create structured audit records without passwords or unnecessary medical details. Backups are encrypted, access-controlled and periodically restored in the acceptance environment. Restoration evidence includes completion time and record comparison.

7 Sequence Design

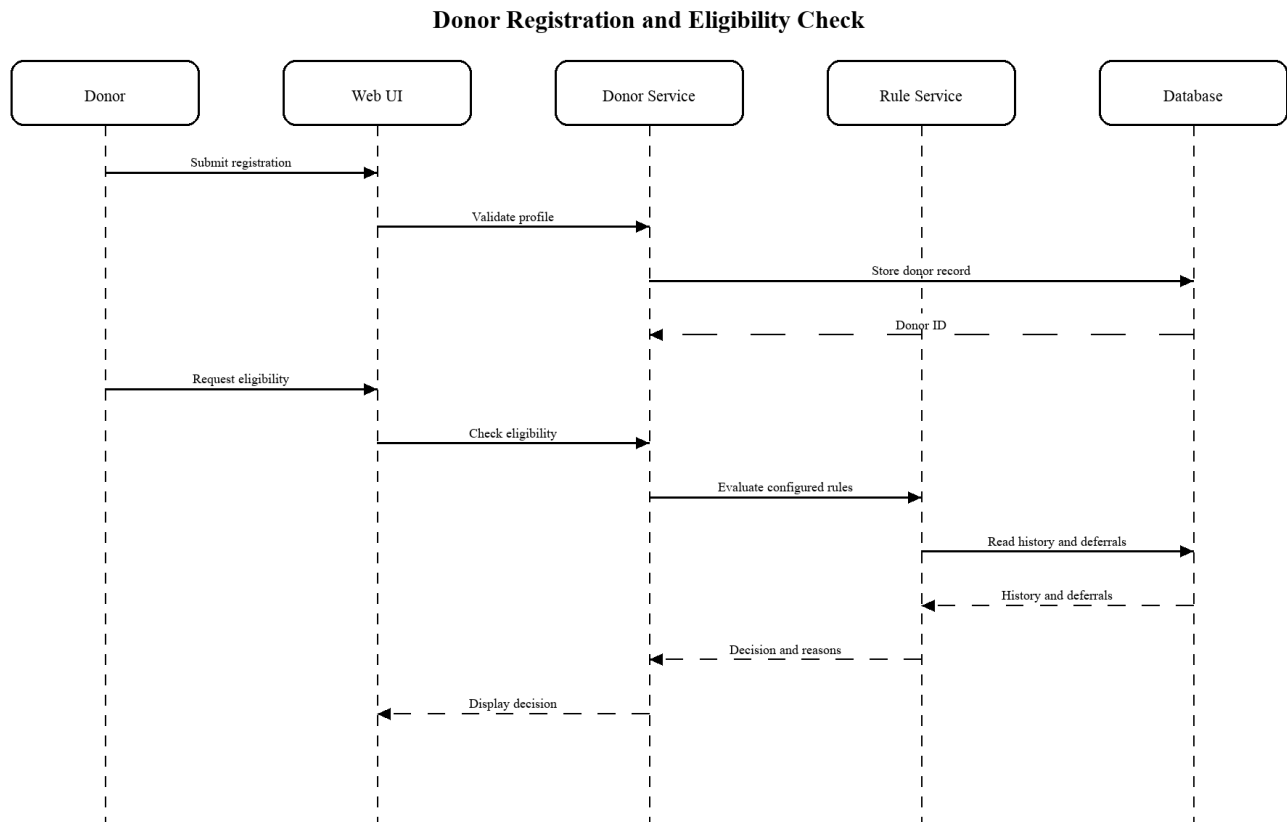


Figure 4 Sequence diagram for registration and eligibility

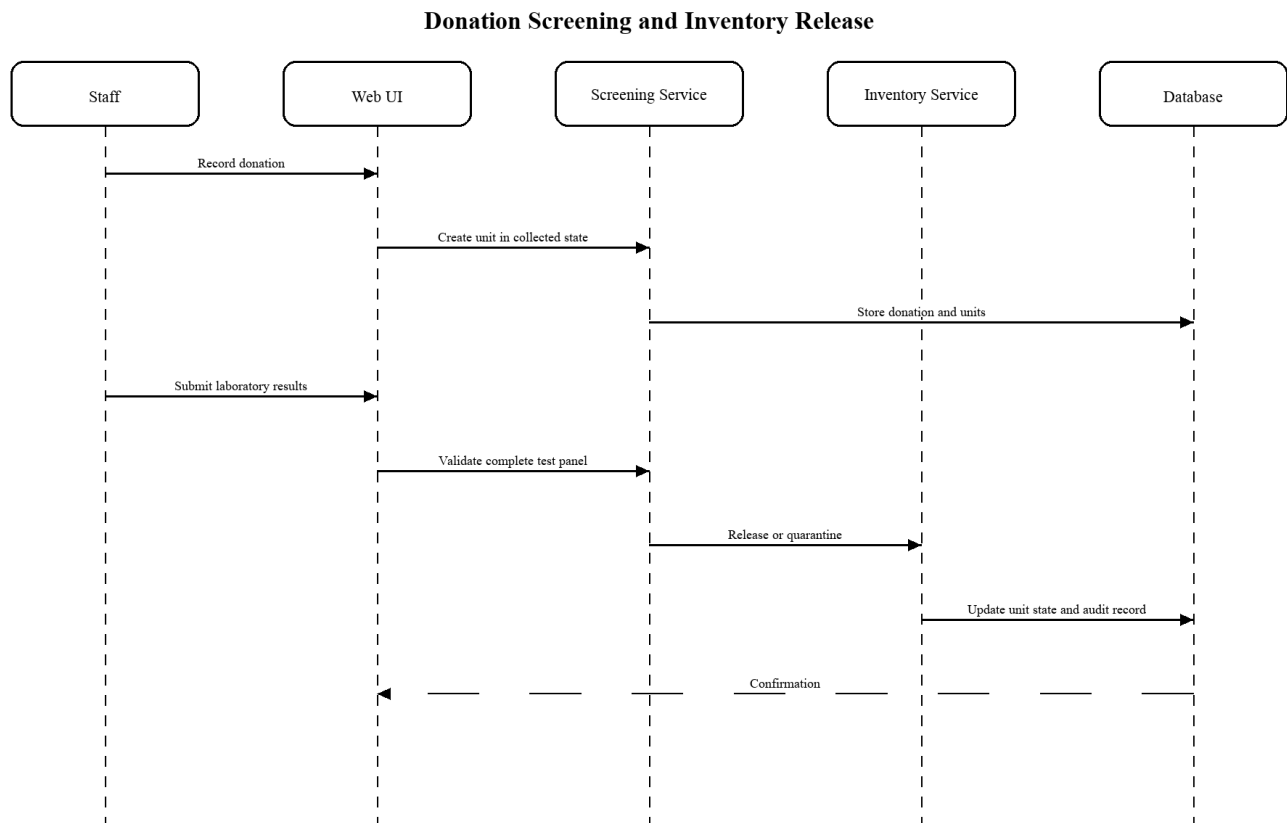


Figure 5 Sequence diagram for screening and inventory release

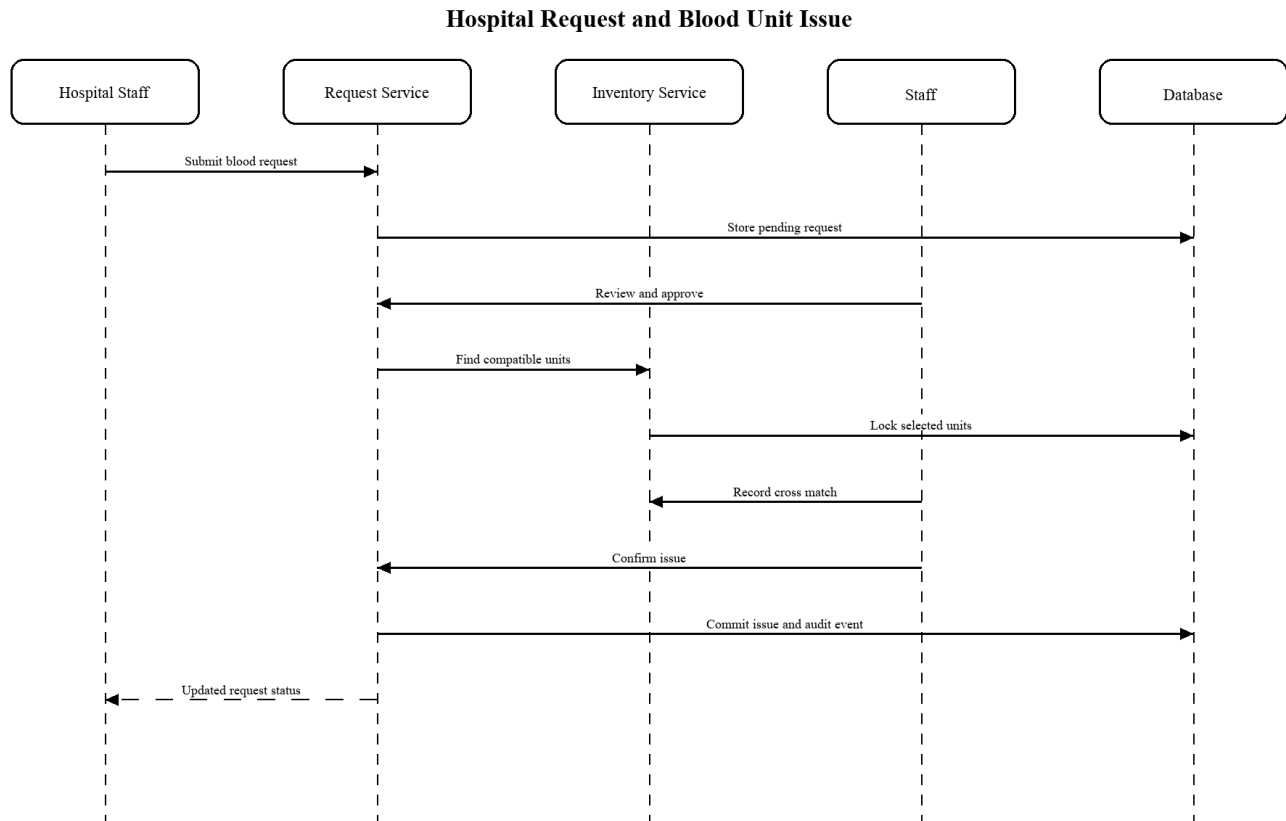


Figure 6 Sequence diagram for request and issue

8 API Design

8.1 API Conventions

- The base path is /api/v1 and payloads use JSON.
- Authentication uses the same secure session as the web application for the first release.
- State-changing requests require CSRF protection and a valid authorized session.
- Validation errors use a stable error code and field-level details.
- Reservation and issue endpoints accept an idempotency key to reduce duplicate processing.

8.2 Endpoint Summary

Table 22 API endpoint summary

Method	Path	Role	Purpose	Responses
POST	/api/v1/auth/login	Public	Authenticate an account and create a session	200, 400, 401, 423
POST	/api/v1/donors	Public	Register a donor profile	201, 400, 409
GET	/api/v1/donors/{id}/eligibility	Donor or Staff	Return eligibility decision and reasons	200, 403, 404
POST	/api/v1/appointments	Donor	Create an appointment	201, 400, 409
PATCH	/api/v1/appointments/{id}	Donor or Staff	Reschedule or cancel an appointment	200, 400, 403, 409
POST	/api/v1/donations	Staff	Record a donation and component units	201, 400, 403, 409
POST	/api/v1/units/{id}/lab-results	Staff	Record a screening result	201, 400, 403, 409
GET	/api/v1/inventory	Staff or Hospital Staff	Search authorized stock information	200, 400, 403
POST	/api/v1/requests	Hospital Staff	Create a blood request	201, 400, 403
PATCH	/api/v1/requests/{id}/status	Staff	Approve, reject or cancel a request	200, 400, 403, 409
POST	/api/v1/requests/{id}/reserve	Staff	Reserve compatible units atomically	200, 400, 403, 409
POST	/api/v1/issues	Staff	Issue cross-matched units	201, 400, 403, 409
GET	/api/v1/reports/inventory	Admin or Staff	Generate an inventory report	200, 400, 403

8.3 Standard Error Body

An API error contains code, message, field_errors when relevant, correlation_id and timestamp. Messages do not expose stack traces, database details, password status or the existence of an unregistered account.

9 Error Handling

Condition	System response	HTTP or UI result	Audit or log
Invalid field input	Reject before service execution	400 with field errors	Application log for repeated abuse only
Unauthenticated access	Do not execute operation	401 or login redirect	Authentication event
Unauthorized access	Do not disclose protected record	403 with generic message	Security audit event
Missing record	Return stable not-found response	404	Normal request log
State or concurrency conflict	Rollback and return current state	409	Operational audit event
Email failure	Record failure and schedule bounded retry	Primary transaction remains committed	Integration log
Database failure	Rollback transaction and show correlation ID	500 with generic message	Error log and administrator alert

9.1 Exception Handling Rules

- Domain exceptions represent validation, authorization, state-transition and conflict errors.
- Views translate known exceptions into stable user messages and status codes.
- Unexpected exceptions are logged once with a correlation ID and produce a generic response.
- Transactions roll back whenever one step of reservation or issue fails.
- No exception message contains a password, secret, full medical history or database query.

10 Design Traceability

Requirement group	Primary components	Design evidence
FR-01 to FR-08	Donor and Appointment Service	Use cases UC-01, state validation and Figure 4
FR-09 to FR-13	Donation and Screening Service	UC-02, UC-03 and Figure 5
FR-14 to FR-24	Inventory and Request Service	State tables, UC-04, UC-05 and Figure 6
FR-25 to FR-30	Authentication, Administration, Reporting and Audit	Role matrix, API and error-handling design
NFR-01 to NFR-12	All layers and deployment	Performance limits, constraints, backup and compatibility design
SEC-01 to SEC-10	Security architecture	Sections 6.1 to 6.5 and TC-S01 to TC-S03

11 Design Decisions and Limitations

- A modular monolith avoids the operational cost of distributed services while retaining service boundaries.
- PostgreSQL is selected to remove the earlier unresolved database choice and to support constraints and transactions.
- Clinical analyser integration, barcode printing and national identity integration remain future work.
- The design requires professional approval of eligibility, shelf-life and retention configuration before operational use.

References

- [1] ISO IEC IEEE 29148 2018, Systems and software engineering, Life cycle processes, Requirements engineering.
- [2] IEEE 829 2008, Standard for Software and System Test Documentation, used as the course-aligned document structure. The standard has been superseded by the ISO IEC IEEE 29119 series.
- [3] IEEE 1016 2009, Standard for Information Technology, Systems Design, Software Design Descriptions.
- [4] Ministry of Health and Family Welfare, Government of India, National Standards for Blood Centres, Annexure 4 Criteria for Selection of Blood Donor.
- [5] Central Drugs Standard Control Organization, Drugs and Cosmetics Act 1940 and Rules 1945, Schedule F provisions for blood banks.
- [6] eRaktKosh, Ministry of Health and Family Welfare, overview of centralized donor, testing, inventory and notification functions.
- [7] Django Software Foundation, Django documentation for authentication, security and database transactions.

Appendix A Deliverables Compliance Checklist

Deliverable requirement	Location in submission	Status
SRS in IEEE-aligned format	Part A Sections 1 to 8	Complete
Introduction and descriptive SRS sections	Part A Sections 1 to 3	Complete
Clear functional and non-functional requirements	Part A Section 4	Complete
At least one UML use-case diagram	Part A Figure 1	Complete
At least two security objectives and requirements	Part A Section 6	Complete
Test Plan in IEEE-aligned format	Part B Sections 1 to 8	Complete
Test Plan Sections 3 4 and 5	Part B Sections 3 to 5	Complete
Security validation in Section 5.1	Part B Section 5.1	Complete
Traceability with SRS	Part A Section 8 and Part B Section 7	Complete
Component diagram and description	Part C Figure 2 and Section 4	Complete
Architecture pattern	Part C Section 3.1	Complete
Architecture traceability	Part C Section 10	Complete
Security architecture	Part C Section 6	Complete
At least two UML sequence diagrams	Part C Figures 4 to 6	Complete
API design	Part C Section 8	Complete
Error handling	Part C Section 9	Complete
At least seven to ten functional and non-functional test cases	Part B Section 6 contains nineteen cases	Complete